

Bridging measurements and simulations: A data assimilation approach to real time analysis

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Abstract

Accurate simulation of wall-bounded turbulent flows remains challenging due to the high computational cost of resolving near-wall dynamics. Experimental techniques provide valuable data in this region, but are often limited by resolution. A promising strategy is to combine numerical simulations with wall-based measurements through data assimilation (DA) aiming to real-time .

In this work, we investigate which types of wall measurements are most effective for improving flow reconstruction while reducing computational cost. A sudden acceleration boundary-layer configuration, governed by the Navier–Stokes and energy equations, is used as the test case. High-resolution simulations provide the ground truth, from which synthetic wall observations (temperature, pressure, shear stress) are extracted and assimilated into a coarser simulation.

Four DA methods—nudging, Kalman Filter (KF), Ensemble Kalman Filter (EnKF), and 4D-Var—are benchmarked in terms of accuracy and efficiency. All methods achieve similar accuracy (RMSE (root mean square error) ≈ 0.13), but with significant differences in cost. Nudging and KF provide the best balance, with nudging requiring less than 1 ms per update. EnKF enables multivariable assimilation at moderate cost, while 4D-Var achieves high accuracy but at the highest computational expense. The results demonstrate the potential of lightweight DA approaches to integrate sparse wall-based experimental measurements into computational fluid dynamics workflows, bridging the gap between simulations and experiments.

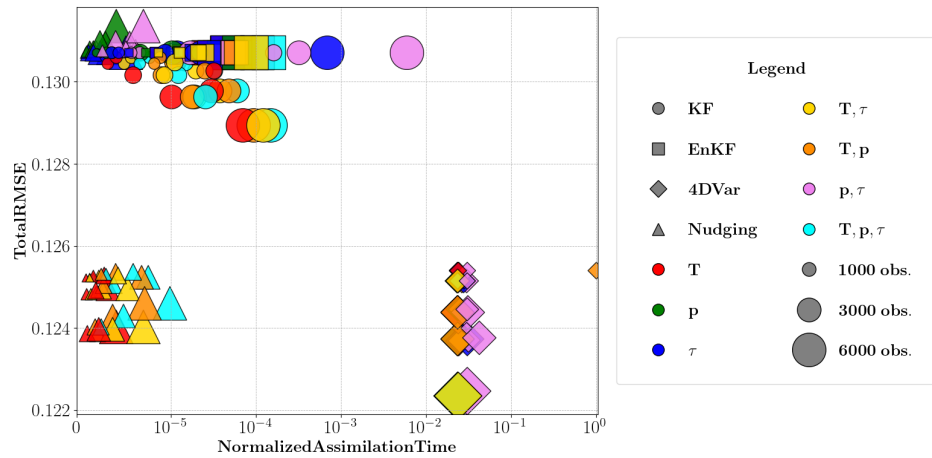


Figure 1: Accuracy (Total RMSE) versus normalized assimilation time for different DA methods. Marker shape = method, color = type of wall observation, size = number of observations.